7 Results and Discussion

7.1 Evaluation Protocol

All results in this section use one shared pipeline and one backtest, so that the models are directly comparable. Three properties of the protocol matter for reading the numbers.

1. **Three-way chronological split with purging.** The 1,030,551 aligned snapshots are split 70/15/15 into training (04-07 11:34 to 04-15 20:00 UTC), validation (to 04-17 14:57) and test (to 04-19 09:54). A row is used as a target only if its whole label window $[t, t+h]$ lies inside its own split. Early stopping uses the *validation* loss.

2. **Trading rules are chosen on validation only.** Every tunable trading parameter (the confidence threshold θ and the hysteresis enter/exit levels) is selected by validation Sharpe and then applied once to the test split. An earlier draft of this project selected θ on the test split; those numbers are not reported here.

3. **Fee-aware, non-overlapping backtest.** A position $q_t \in \{-1, 0, +1\}$ (full capital, no leverage) is held for exactly h seconds and earns the mid-price return $r_{t,h}$. Steps do not overlap. The net step return is $q_t r_{t,h} - c\,|q_t - q_{t-1}|$ with $c = 1.5$ bp, so a reversal pays twice, and wealth compounds multiplicatively. Execution is at the mid-price of the same snapshot the features come from. Section 7.6 relaxes this assumption.

Three trading rules map the predicted class probabilities $(p_\downarrow, p_0, p_\uparrow)$ to positions: *argmax* trades every prediction; *threshold* goes long if $p_\uparrow > \theta$, short if $p_\downarrow > \theta$, and is otherwise flat; *hysteresis* opens a position when its class probability exceeds an *enter* level, keeps it while that probability stays above a lower *exit* level, and reverses directly on a strong opposite signal. A LightGBM classifier on the current second's features only serves as a non-deep baseline.

7.2 Classification Performance

Table 1 reports test classification metrics at the 10-second horizon. All four models reach 41–44% accuracy, which is *no better than always predicting Flat* (43.1% of test labels). Accuracy is therefore not informative here. Directional F1, the macro F1 over the Down and Up classes, does show a weak but consistent directional signal of 0.44–0.47. The Transformer has the highest directional F1 (0.466), but the gap to the other models is small: LightGBM, which sees only the current second, reaches 0.465. The sequence architectures add little classification skill beyond the hand-crafted features.

Table 1: Test classification metrics, 10-second horizon, main split (154,544 test targets). DirF1 = macro F1 over Down and Up. Always predicting Flat gives an accuracy of 0.431.

	Full (38F)		No PCA (88F)		No Market (23F)	
Model	Acc	DirF1	Acc	DirF1	Acc	DirF1
Residual CNN	0.445	0.454	0.444	0.448	0.443	0.437
BiLSTM	0.434	0.447	0.434	0.449	0.436	0.445
Transformer	0.437	**0.466**	0.434	0.464	0.436	0.456
LightGBM baseline	0.409	0.465	0.408	0.461	0.413	0.444

Across feature sets, removing the market-state block (No Market) lowers directional F1 for every model (by 0.002–0.020), while replacing PCA factors with the raw 60 LOB columns (No PCA) changes it by less than 0.007. Cross-asset features help slightly; PCA compression neither helps nor hurts.

7.3 Trading Performance on the Main Split

Table 2 reports the test backtest at the 10-second horizon. Buy-and-hold loses 6.5% over the test window, which contains the 18 April sell-off (maximum drawdown −14.8%).

Table 2: Test returns at the 10-second horizon (15,457 non-overlapping steps, 1.5 bp fee). θ and the hysteresis levels are chosen on validation. Turnover = total $\sum_t |q_t - q_{t-1}|$. Buy-and-hold: −6.52%.

		Argmax	Threshold		Hysteresis	
Model	Features	Return	Return	Turnover	Return	Turnover
Residual CNN	Full	−40.1%	+0.72%	28	**+26.8**%	2,249
Residual CNN	No PCA	−40.3%	−0.35%	42	–	–
Residual CNN	No Market	−42.2%	−0.71%	62	–	–
BiLSTM	Full	−48.6%	−0.16%	96	+4.5%	428
BiLSTM	No PCA	−46.9%	+0.27%	22	–	–
BiLSTM	No Market	−46.5%	−0.12%	54	–	–
Transformer	Full	−42.8%	+0.32%	152	+3.5%	284
Transformer	No PCA	−39.7%	−0.06%	24	–	–
Transformer	No Market	−42.5%	+0.05%	30	–	–
LightGBM	Full	−39.3%	+0.08%	20	+6.3%	974
LightGBM	No PCA	−42.6%	−0.21%	280	+17.1%	1,922
LightGBM	No Market	−42.5%	−0.34%	114	−0.9%	1,039

Argmax. Trading every prediction loses 39–49% for every model and feature set. Even with a non-trivial directional F1, the expected move per 10-second step is far smaller than the 1.5 bp paid on each position change.

Threshold. A confidence threshold chosen on validation ($\theta = 0.55$–0.85) almost stops the models from trading: 20–280 position changes over 1.8 days, with test returns between −0.7% and +0.7%. With so few trades these results are not distinguishable from zero. In particular, the earlier draft's CNN result of +2.7% (Sharpe 31) does not survive once θ is selected out of sample.

Hysteresis. Separating the entry and exit levels changes the picture. The model now stays in a position through small dips in confidence instead of paying to close and reopen it. Five of the six hysteresis runs turn positive. The CNN earns +26.8% (Sharpe 62, maximum drawdown −6.4%). On this split, the result is not a single lucky configuration. Both long positions (+15.1%) and short positions (+22.9%) contribute, before −14.1% of fees. Each of the three test days is positive (+3.6%, +16.9%, +3.5%). Every neighbouring (enter, exit) pair within ±0.05 also stays positive (+14.1% to +49.0%). The LSTM (+3.0% to +14.7% over the same neighbourhood) and the Transformer (+0.0% to +11.9%) are positive on every test day as well, with far less trading.

7.4 Effect of the Prediction Horizon

A natural way to beat the fee is to predict further ahead, since price moves grow with the horizon while the fee does not. We rebuilt the labels for $h \in \{30, 60, 300\}$ seconds, with the Up/Down threshold raised to the round-trip cost of 3 bp, and retrained every model on the Full feature set (Table 3).

Table 3: Directional F1 and hysteresis test return by prediction horizon (Full feature set, main split). "Flat" = share of Flat labels in the test set, which equals the accuracy of always predicting Flat. Buy-and-hold $\approx -6.6\%$ at every horizon.

		CNN		BiLSTM		Transformer		LightGBM	
h	Flat	DirF1	Hyst.	DirF1	Hyst.	DirF1	Hyst.	DirF1	Hyst.
10 s	0.431	0.454	+26.8%	0.447	+4.5%	0.466	+3.5%	0.465	+6.3%
30 s	0.359	0.413	+0.1%	0.426	+0.7%	0.437	+1.0%	0.446	+13.3%
60 s	0.244	0.436	+8.3%	0.445	+4.0%	0.416	−0.3%	0.466	+8.6%
300 s	0.107	0.420	−4.4%	0.447	−3.9%	0.372	−12.3%	0.476	−2.4%

Longer horizons do not help. Accuracy rises above the Flat baseline only because Flat labels become rare, and directional F1 stays at 0.37–0.48 at every horizon. The apparent gains at 30 and 60 seconds come mostly from short positions during the 18 April sell-off, and they are fragile. For LightGBM at 30 seconds, for example, the test return over the neighbouring (enter, exit) pairs ranges from −12.1% to +14.9%. At 300 seconds every hysteresis run loses money. LightGBM matches or exceeds the deep models' directional F1 at every horizon.

7.5 Walk-Forward Evaluation

The main split has a single 1.8-day test window. Many configurations were compared on it, so its best result is optimistically selected. To test whether the 10-second hysteresis result generalises, we re-ran the CNN and LightGBM on rolling folds. Each fold trains on 5 days, validates on the next day and tests on the day after, and the window then moves forward one day, giving six disjoint test days from 13 to 18 April. Within each fold, the PCA basis is refitted on that fold's training days, the hysteresis levels are chosen on its validation day, and the CNN is trained with three seeds plus a seed-averaged ensemble. The first four test days were not used to choose this configuration, so they are genuinely out of sample.

Table 4: Walk-forward test-day returns (10-second horizon, hysteresis, 1.5 bp fee). Each column is one fold; a "day" starts at 11:34 UTC. CNN single seeds: mean of the three seeds on each day (individual runs range from +1.0% to +35.3%).

Test day	04-13	04-14	04-15	04-16	04-17	04-18	Mean
CNN, 3-seed ensemble	+4.2%	+3.1%	+9.8%	+9.0%	+6.2%	+10.0%	+7.0%
CNN, single-seed mean	+2.9%	+2.7%	+3.6%	+6.7%	+20.6%	+9.8%	+7.7%
LightGBM, No PCA	+3.2%	+8.9%	+7.9%	+5.9%	−0.8%	+11.3%	+6.1%
LightGBM, Full	−2.2%	+1.0%	+12.4%	+9.3%	−2.7%	+10.8%	+4.8%
Buy-and-hold	+2.3%	−3.0%	−3.0%	+0.7%	−10.9%	+4.5%	−1.6%

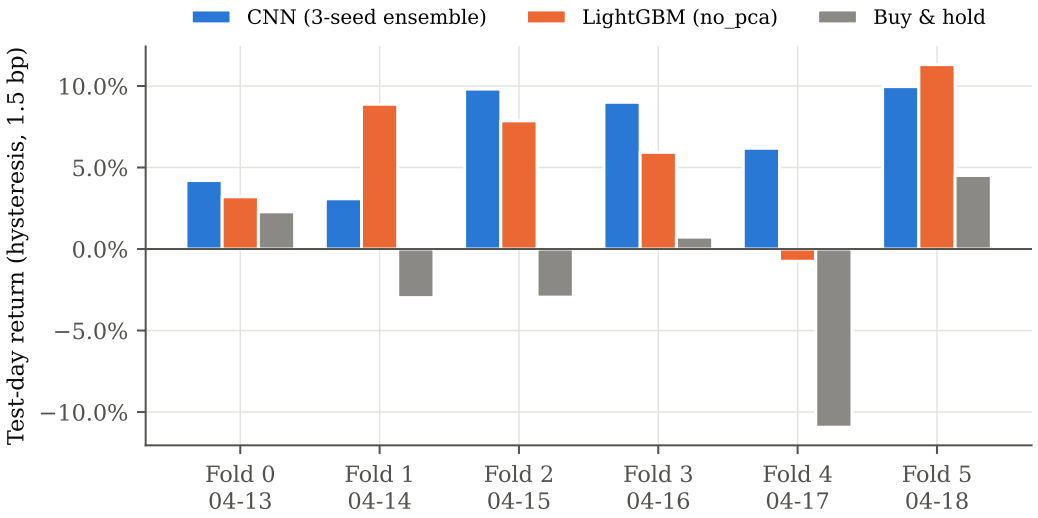


Figure 1: Walk-forward test-day returns of the hysteresis strategy against buy-and-hold.

Table 4 and Figure 1 show that the result is remarkably stable *under the backtest's assumptions*. All 18 single-seed CNN runs and all six ensemble days are positive, including the four test days before the sell-off, which were not used to choose the configuration. Compounded over the six days, the CNN ensemble returns +50.1% against −9.7% for buy-and-hold. Seed averaging lowers the variance: single seeds range from +1.0% to +35.3%, the ensemble from +3.1% to +10.0%. LightGBM is profitable on 5 of 6 days with No PCA features and on 4 of 6 with Full features. Directional F1 stays at 0.41–0.47 in every fold, the same level as on the main split.

7.6 Reality Checks: Execution Delay and Data Alignment

Execution delay. The backtest assumes we trade at the mid-price of the very snapshot from which the features were computed. To test this, we keep every trained model and every validation-chosen threshold fixed and decide the position for the step starting at second t from the prediction made at second $t - d$. This uses only the saved predictions, so no retraining is involved. Figure 2 and Table 5 show the result.

Table 5: Hysteresis test return with execution delayed by d seconds (10-second horizon, same models and thresholds). Walk-forward rows: mean over six test days (share of positive days in brackets).

	$d = 0$	$d = 1\,\mathrm{s}$	$d = 2\,\mathrm{s}$	$d = 3\,\mathrm{s}$	$d = 5\,\mathrm{s}$
Walk-forward (mean per test day)					
CNN ensemble	+7.0% (6/6)	+0.9% (4/6)	−1.9% (3/6)	−4.0% (2/6)	−6.8% (0/6)
CNN single seeds	+7.7%	+2.3%	−1.0%	−2.9%	−4.9%
LightGBM (both sets)	+4.9%	−3.6%	−4.2%	−6.3%	−9.0%
Main split (test 04-17 to 04-19)					
CNN	+26.8%	+4.7%	−5.2%	−9.4%	−10.1%
BiLSTM	+4.5%	+1.4%	+0.4%	−0.5%	−2.1%
Transformer	+3.5%	+0.9%	+0.0%	−1.0%	−1.6%
LightGBM, No PCA	+17.1%	−2.5%	−9.8%	−13.4%	−12.3%

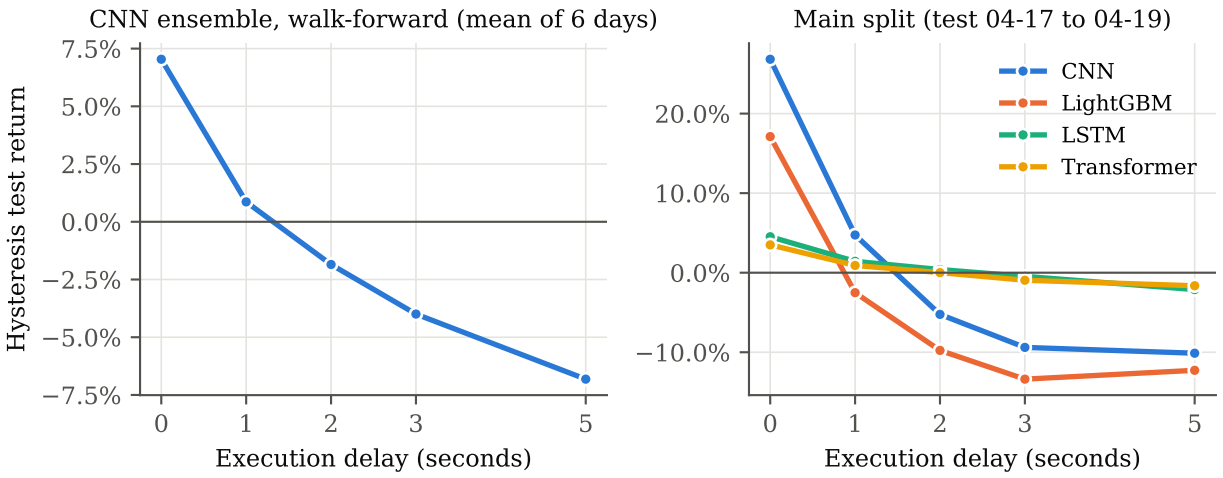


Figure 2: Hysteresis test return as a function of execution delay. Left: CNN ensemble, mean over the six walk-forward test days. Right: all models on the main split.

For every model, most of the profit disappears with a one-second delay, and it turns negative at two to three seconds. The models do anticipate the next move of the mid-price, but that move happens within roughly one second of the snapshot. The LSTM and Transformer lose less when delayed only because their validation-chosen entry levels are higher, so they trade less. Their profit decays in the same way.

BTC–ETH timestamp alignment. BTC and ETH snapshots are joined on the floored second. Within a matched second (first 300,000 snapshots), the ETH snapshot is recorded *later* than the BTC snapshot 24.5% of the time, by up to 0.74 s (19.7% of seconds are more than 0.1 s later). In those seconds, the ETH-based market features contain information from slightly after the BTC mid-price they are compared with, which is a small look-ahead in every feature set that includes market features. A delay of one second or more removes this leak as well, which is a further reason to treat the delayed numbers as the realistic ones.

7.7 Discussion

The results support three conclusions.

1. **The models learn genuine but very short-lived order-book information.** Directional F1 of 0.41–0.47 is stable across architectures, feature sets, seeds and six walk-forward folds. Under instantaneous execution, a hysteresis rule turns this signal into profits that are consistent across days and robust to the choice of thresholds.

2. **The signal is not tradable at 1-second resolution after a 1.5 bp fee.** Its value is concentrated in the first second after each snapshot. With any realistic latency, no model, feature set, horizon or trading rule in this study has a positive expected return. Capturing it would require sub-second execution at the mid-price, with no queueing or adverse selection, which this data cannot evaluate.

3. **Architecture matters less than evaluation design.** The three deep models and a LightGBM baseline that sees only the current second reach similar directional F1. The choices with

the largest effect on reported returns are methodological: selecting thresholds out of sample (which removes the earlier +2.7% CNN result), separating entry from exit, and modelling execution delay (which removes all remaining profit).

Limitations. The data covers only 12 days in April 2021, and all test days fall in one week that contains a single large sell-off. The backtest fills at the mid-price with a flat fee, ignores market impact and borrowing costs for short positions, and uses full-capital positions. The 1.5 bp fee is lower than retail taker fees at the time. Only the CNN and LightGBM were evaluated walk-forward; the LSTM and Transformer were checked for execution delay on the main split only.